

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary - Winter 23

Course: B. Tech. Branch : Computer Engineering/Computer Science & Engineering

Subject Code & Name: BTCOE604 (B)_Y23 Internet of Things

Semester :VI

Max Marks: 60

Date:23-01-24

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. Draw neat diagram wherever necessary.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- | | | |
|---|-------------|----|
| A) What is IoT? Explain its characteristics and application of IoT. | Application | 06 |
| B) What are the biggest challenges for IoT adoption? | Understand | 06 |
| C) Explain with neat diagram different IoT communication models. | Remember | 06 |

Q.2 Solve Any Two of the following.

- | | | |
|---|------------|----|
| A) Explain communication criteria for an IoT platforms. | Understand | 06 |
| B) What is actuator? Give classification of actuators. | Analysis | 06 |
| C) Explain leading types of IoT wireless technologies. | Understand | 06 |

Q.3 Solve Any Two of the following.

- | | | |
|---|------------|----|
| A) List and explain key advantages of Internet protocol (IP). | Understand | 06 |
| B) Describe Message Queuing Telemetry Transport Protocol. | Understand | 06 |
| C) Explain supervisory control and data acquisition. | Understand | 06 |

Q.4 Solve Any Two of the following.

- | | | |
|--|------------|----|
| A) What is machine learning? Explain types of machine learning categories. | Remember | 06 |
| B) What is Edge analytics? Describe Edge Analytics Core Functions. | Understand | 06 |
| C) Explain Massively Parallel Processing Shared-Nothing Architecture. | Understand | 06 |

Q.5 Solve Any Two of the following.

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|--|-------------|----|
| A) Explain smart city IoT architecture with a neat diagram. | Understand | 06 |
| B) Explain smart city security architecture with a neat diagram. | Understand | 06 |
| C) Discuss an IoT Strategy for Smarter Cities. | Application | 06 |

*** End ***